CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 17: Circles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. An arc subtends 140° at the centre of a circle. Find the angle it subtends at a point on the remaining part of the circle.

2. An arc subtends 35° at a point on the major arc. Find the angle it subtends at the centre.

3. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 62^\circ$. Find $\angle BAC$.

4. O is the centre of a circle and $\angle OAB = 25^\circ$ for a chord AB. Find the angle that AB subtends at a point on the major arc.

5. For the chord in the previous question, find the angle it subtends at a point on the minor arc.

6. $ABCD$ is a cyclic quadrilateral with $\angle B = 95^\circ$ and $\angle C = 80^\circ$. Find $\angle A$ and $\angle D$.

7. In a cyclic quadrilateral $PQRS$, $\angle P = 3x^\circ$ and $\angle R = 2x^\circ$. Find x.

8. $ABCD$ is a cyclic quadrilateral and DC is produced to E. If $\angle BAD = 68^\circ$, find $\angle BCE$.

9. Points P and Q lie on the same side of a chord AB of a circle. If $\angle APB = 48^\circ$, find $\angle AQB$ and give the reason.

10. Prove that a cyclic parallelogram is a rectangle.

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Answer Key

1. 70°
2. 70°
3. 28°
4. $\angle AOB = 130^\circ; 65^\circ$
5. $\frac{1}{2}(360^\circ - 130^\circ) = 115^\circ$
6. $\angle A = 100^\circ, \angle D = 85^\circ$
7. $x = 36$
8. 68°
9. 48° (angles in the same segment)
10. Opposite angles of a parallelogram are equal, and in a cyclic quadrilateral they add up to 180°, so each is 90°.

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